Guadalupe Gutierrez Avila A01772447
Maria Paulina Olmos Diaz A01772741
Josue Fernando Almazan Martinez A01772640

① $\int \frac{dx}{x}$

$$\frac{dx}{x} = \frac{1}{x} = \ln|x| + C$$

$$(\ln|x|)\Big|_e^1 = \ln|x| + C(e)$$

$$0 - 1 = -1$$

② $\int_1^e \frac{dx}{x} = \frac{1}{x} = \ln|x|$

$$(\ln(x))\Big|_e^1 = \ln(1) - \ln(e)$$

$$0 - 1 = -1$$

③ $\int (2x^3 + x^2 - x + 4) \cdot dx$

$$2x^3 dx \int x^2 dx \int -x\,dx \int 4dx$$

$$\frac{2x^4}{4} + \frac{x^3}{3} - \frac{x^2}{2} + 4x$$

$$= \frac{1}{2}x^4 + \frac{x^3}{3} - \frac{x^2}{2} + 4x$$

④ $\int_0^1 (2x+1)\cdot dx$

$2x\,dx \int +1\,dx$

$\frac{2x^2}{2} + 1x$

$= \frac{x^2}{2} + x$

$= \frac{(1)^2}{2} + 1 = 1 + 1 = \boxed{2}$

$(0)^2 + 0 = 0$

⑤ $\int \sqrt{5x+7}\cdot dx$

$(5x+7)^{\frac{1}{2}} dx$

$\frac{du}{dx} = 5 \quad du = 5dx \quad = dx = \frac{du}{5}$

$\int u^{\frac{1}{2}} \left(\frac{du}{5}\right)$

$\frac{1}{5}\int u^{\frac{1}{2}}\,du$

$\frac{1}{2}+1 = \frac{1}{2}+\frac{2}{3} = \frac{3}{2} \quad = \frac{1}{5}\left(\frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right)$

$\frac{1}{5}\left(\frac{2}{3}u^{\frac{3}{2}}\right) = \frac{2}{15}u^{\frac{3}{2}}$

$= \frac{2}{15}(5x+7)^{\frac{3}{2}} + C$

$= \boxed{\frac{2}{15\sqrt{(5x+7)^3}} + C}$

6) $\int_0^{\pi} \sin(x) \cdot dx = -\cos(x)\Big|_0^{\pi} = -\cos(0) - (-\cos(\pi)) = -2$ //

7) $\int_1^0 2 \cdot x \cdot (x^2+1)^2 \cdot dx$

$\int_1^0 2x\,(x^4 + 2x^2 + 1)\,dx$

$\int_1^0 (2x^5 + 4x^3 + 2x)\,dx$

$2\int_1^0 x^5 + 4\int_1^0 x^3 + 2\int_1^0 x$

$\frac{2x^6}{6} + \frac{4x^4}{4} + \frac{2x^2}{2} = \frac{x^6}{3} + x^4 + x^2 + C$ //

9) $\int_{-1}^{1} \frac{2x}{\sqrt{x^2-1}} \cdot dx$ $\quad u = x^2 - 1$

$du = 2x \cdot dx$

$du/dx = 2x \cdot dx$

$= \int \frac{du}{\sqrt{u}} = \int u^{-1/2} \cdot du = \frac{u^{1/2}}{\frac{1}{2}} = 2\sqrt{u} + C$ //

$= 2\sqrt{x^2-1} + C$ //

8) $\int_{-1}^{1} (x^3 + x)\,dx = \frac{x^4}{4} + \frac{x^2}{2}\Big|_{-1}^{1} = \frac{(+1)^4}{4} + \frac{(+1)^2}{2} - \left(\frac{(-1)^4}{4} + \frac{(-1)^2}{2}\right) = 0$ //

10) $\int_0^1 x\,(3x^2-1)^6 \cdot dx$ $\quad u = 3x^2 - 1$

$du = 6x \rightarrow \frac{du}{6} = x \cdot dx$

$3(1)^2 - 1 = 2$

$\frac{1}{6}\int_0^1 u^5 \cdot du = \frac{u^6}{36} = \frac{(3x^2-1)^6}{36}\Big|_0^1 = \frac{(3(1)^2-1)^6}{36} - \frac{(3(0)^2-1)^6}{36} = -\frac{71}{4}$ //

$3(0)^2 - 1 = -1$

(11) $\int \frac{\sin x \cdot dx}{1+\cos x}$, $u = 1+\cos x \iff du = -\sin x\, dx$, $-du = \sin(x)\, dx$

$\int_1^2 \frac{1}{u} \cdot -du \iff -\int_1^2 \frac{1}{u}\, du$

$-\ln|u|\Big|_1^2 \iff -\ln|1+\cos x|\Big|_1^2 \iff -\ln|1+\cos 2| - (-\ln 1+\cos(1))$

$= \ln\left|\frac{1+\cos(1)}{1+\cos(2)}\right| = 0.97008$ Esto es radianes

(12) $\int (5e^x + 1)^2 dx \iff \int 5e^{2x} dx + \int 10e^x dx + \int dx$

$u = 2x$ $\quad \frac{25}{2}\int e^u du + 10\int e^x dx + x + C$

$du = 2dx$

$\frac{5du}{2} = 5dx$ $\quad \frac{25e^u}{2} + C + 10e^x + C + x + C$

$\frac{25e^{2x}}{2} + 10e^x + x + C$

(13) $\int_0^5 (3x+5)^2 dx$, $u = 3x+5$, $du = 3dx \iff \frac{du}{3} = dx$

$\int_0^5 u^2 \frac{du}{3} \iff \frac{1}{3}\int_0^5 u^2 du \iff \frac{u^8}{24}\Big|_5^{20}$

$\frac{20^8}{24} - \frac{20^5}{24} = 1,066,533,333$

(14) $\int \sqrt{x} \cdot (x^{3/2}+1)^8 \cdot dx$, $u = x^{3/2}+1$, $du = \frac{3\sqrt{x}}{2} dx$

$\int u^8 \frac{2du}{3} \iff \frac{2}{3}\int u^8 du$ $\quad \frac{2du}{3} = \sqrt{x}\, dx$

$\frac{2u^9}{27} + C \iff \frac{2(x^{3/2}+1)^8}{27} + C$